

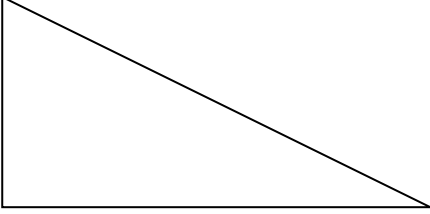


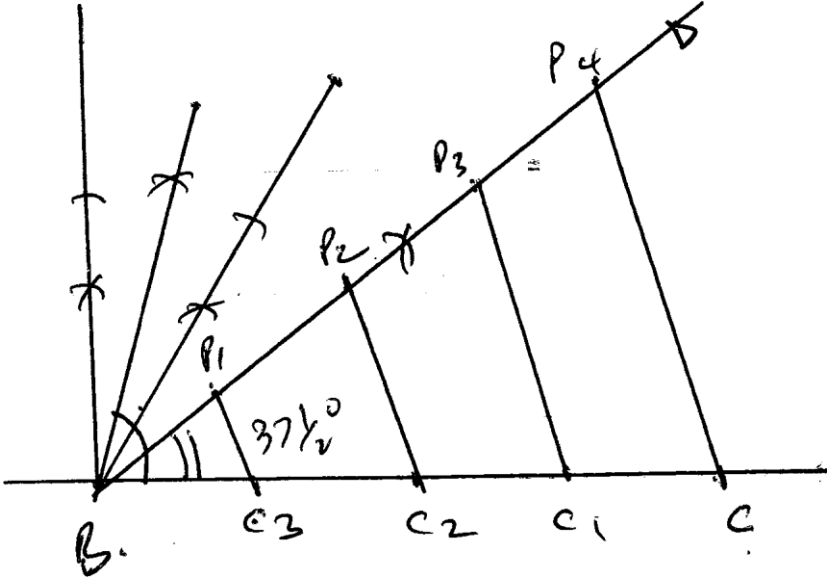
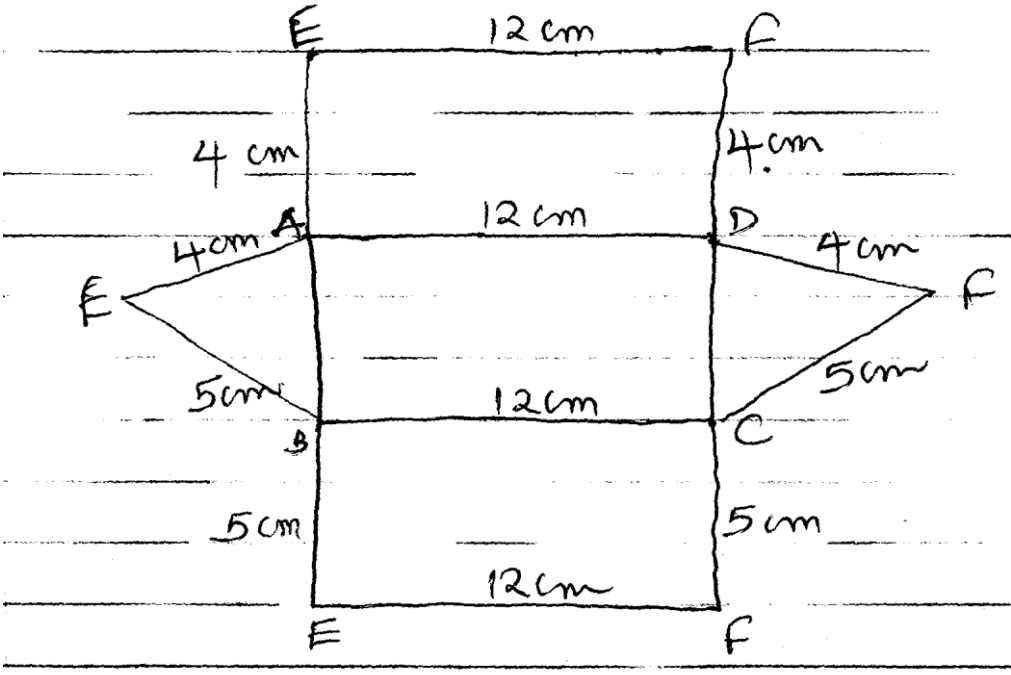
MATHEMATICS 121/1
MARKING SCHEME

No. .	SECTION 1																															
1.	$\frac{3}{4} + 1\frac{7}{7} \div \frac{4}{7} \times \frac{7}{3}$ $= \frac{3}{4} + \frac{9}{7} = \frac{21 + 36}{28} = \frac{57}{28}$ $\left(\frac{80 - 35}{56}\right) \times \frac{2}{3}$ $= \cancel{45}/\cancel{56} \times \cancel{2}/\cancel{7}$ $= \cancel{15}/\cancel{28}$ $= \frac{57}{28} \times \frac{28}{15} = \frac{19}{5}$ $= 3\cancel{4}/\cancel{5}$		M 1 A1																													
2.	$\frac{7}{12} \times \frac{13}{18} = \frac{91}{216}$ $= \frac{91}{216} \times \frac{5}{18} = \frac{91 + 60}{216} = \frac{151}{216}$ $\left(\frac{5}{18} \div \frac{65}{216}\right) 100$ $= \frac{5}{18} \times \frac{216}{65} \times 100 = \frac{1200}{13}$ $= 92\frac{4}{13} \%$		M 1 A1																													
3	L.C. M $2^3 \times 5^2 \times 11$ <table><tr><td></td><td>1400</td></tr><tr><td>2</td><td>700</td></tr><tr><td>2</td><td>350</td></tr><tr><td>2</td><td>175</td></tr><tr><td>5</td><td>35</td></tr><tr><td>5</td><td>7</td></tr><tr><td>7</td><td>1</td></tr></table> <table><tr><td></td><td>1960</td></tr><tr><td>2</td><td>980</td></tr><tr><td>2</td><td>490</td></tr><tr><td>2</td><td>245</td></tr><tr><td>5</td><td>47</td></tr><tr><td>7</td><td>7</td></tr><tr><td>7</td><td>1</td></tr></table>			1400	2	700	2	350	2	175	5	35	5	7	7	1		1960	2	980	2	490	2	245	5	47	7	7	7	1	M 1	
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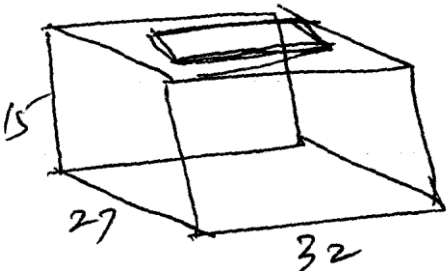
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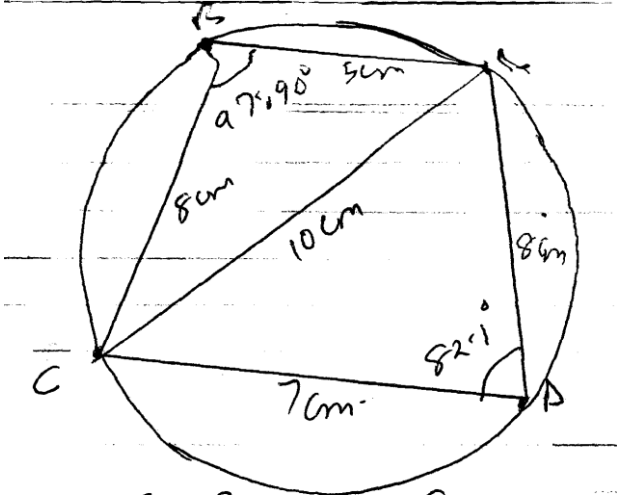
6.	$\frac{4x(3x-4)}{20-15x+4x-3x^2} = \frac{4x(3x-4)}{5(4-3x)+x(4-3x)}$ $= \frac{4x(3x-4)}{(5+x)(4-3x)}$ $= \frac{-4x(4-3x)}{(5+x)(4-3x)}$ $= \frac{-4x}{5+x}$	M 1	
		M 1	
		A1	
7.	 $y = \sqrt{9-4x}$ $= \sqrt{5}$ $\tan^2 \theta + \cos^2 \theta$ $\left(\frac{4}{\sqrt{5}}\right)^2 + \left(\frac{\sqrt{5}}{5}\right)^2$ $= \frac{16}{5} + \frac{1}{5} = \frac{17}{5}$	B1	
		M 1	
		A1	
8.	$8x(2^y)^y = 6(2^y) - 1$ $= 8x(2^y)^2 = 6x(2^y) - 1$ Let $2^y = x$ $\therefore 8x^2 = 6x - 1$ $\Rightarrow 8x^2 - 6x + 1 = 0$ $\Rightarrow 8x^2 - 4x - 2x + 1 = 0$ $4x(2x-1) - 1(2x-1) = 0$ $\Rightarrow (4x-1)(2x-1) = 0$ $\Rightarrow \therefore x = \frac{1}{2} \text{ or } \frac{1}{4}$ $2^y = \frac{1}{2} = 2^{-1} \text{ or } 2^y = \frac{1}{4} = 2^{-2}$ $\therefore y = -1$ or $y = -2$	M 1	
		M 1	
		M 1	
		A1	
9.		B1	$\sqrt{\text{angle}}$

		B1 B1 B1	37.5° $\checkmark$ Division of BD $\checkmark$ Division of BC
10.		B1 B1 2	
11.	$x + \frac{x-36}{3} = 180^\circ$ $\Rightarrow 3x + x - 36 = 540$ $4x = 576$ $\Rightarrow x = 144^\circ$ $\text{Exterior angle} = \frac{144 - 36}{3} = \frac{108}{3} = 36^\circ$ $\therefore \text{No of sides of the polygon}$ $n = \frac{360}{36} = 10 \text{ sides}$	M 1 B1 B1 3	



12.	$AC^2 = 85^2 - 75^2$ $AC = \sqrt{85^2 - 75^2} = 40m$ Area of quadrilateral ABCD $\frac{1}{2} \times 40 \times 75 + \sqrt{75(75-60)(75-50)(75-40)}$ $= 1500 + \sqrt{984375}$ $= 2492m^2$ $\text{In hectares} = \frac{2492}{10000} = 0.2492 \text{ hectares}$ $\approx 0.25 \text{ hectares.}$	M 1 M 1 A1	4
13.	 External volume = $(32 \times 27 \times 15) = 12960 \text{ cm}^3$ Internal volume = $(30 \times 25 \times 14) = 10500 \text{ cm}^3$ Volume of material = $(12960 - 10500)$ Metre used = 2460 cm^3 $\text{Density} = \frac{\text{mass}}{\text{volume}} = \frac{1500}{2460}$ 0.6098 g / cm^3	M 1 M 1 A1 3	
14.	$2x > 5x - 3 > -8$ $2x + 3 > 5x - 3$ $-3x > -6$ $x < 2$ $5x - 3 > -8$ $5x > -5$ $x > -1$ $-1 < x < 2$	M 1 M 1	

		A1	
	The integral values are (0,1)	3	
15.	$\text{Remaining amount} = \frac{9600}{1\text{E}} \times \frac{95}{100} \times 125$ $= \text{ksh}1,140,000$ $\text{Amount speed} = \text{ksh} \frac{1140,000 \times 3}{4}$ $= \text{ksh}855,000$ $\text{The Balance} = \text{ksh } 1,140,000 - 855,000$ $= \text{ksh}285,000$ $\text{Amount in dollars} = \frac{285,000}{63.20}$ $= 4509.49 \text{USdollars}$	M 1 M 1 A1	
16	$1.2 \times 10 \times x = 6$ $x = \frac{6}{12}$ $= 0.5$ $3.2 \times 15 \times 0.5 = 24$	M 1 A1 A1	

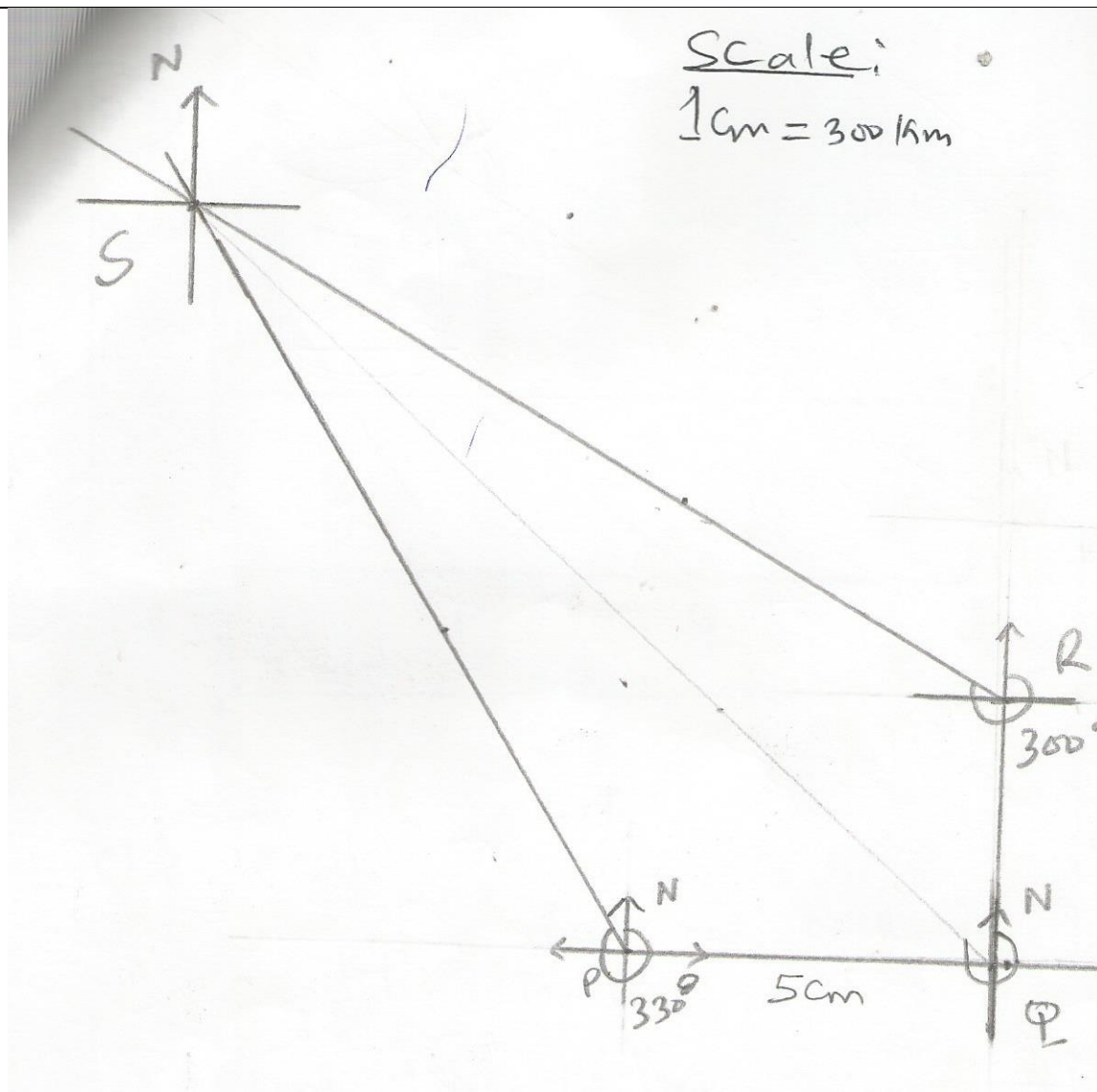
17	 $10^2 = 8^2 + 5^2 - 2 \times 8 \times 5 \cos B$ $\cos B = \frac{89 - 100}{80} = \frac{-11}{80}$ $B = \cos^{-1}\left(\frac{-11}{80}\right) = 97.90^\circ$ (b) $2R = \frac{10}{\sin 97.90}$ $R = \frac{5}{\sin 97.90} = 5.0479 \text{ cm}$ □□□(c) $\frac{10}{\sin 82.1^\circ} = \frac{7}{\sin A}$ $\sin A = \frac{7}{10} \sin 82.1 = 0.6984$ $A = \sin^{-1}(0.6934) = 43.90^\circ$ $\angle COD = 2 \times 43.90^\circ = 87.80^\circ$ $\text{Area} = \frac{87.80}{360} \times \frac{22}{7} \times 5.0479 - \frac{1}{2} \times 5.04329 \sin 87.80$ $= 19.5316 - 12.7313$ $= 6.800 \text{ cm}^2$	M 1 M 1 A1 M 1 A1 M 1 1 M 1 A1	
18	→ (a) (i) $AB = b - a$ ~ ~ → (ii) $CD = -3b + -\frac{1}{3}a$ ~	B1 B1	



	$(b) \text{ CM} = \frac{1}{3} Ka - 3kb$$\sim$$\Rightarrow \text{AM} = -h a + h b$$\sim \quad \quad \sim$$\text{CM} = \text{CA} + \text{AM}$$= \frac{1}{3} K a - 3 b$$\sim \quad \quad \sim$$= a (1 - h) + b(h - 3)$$\sim$$K = 3 - 3h \text{ and } \frac{1}{3} K = 1 - h$ $h = \frac{3}{4}$$k = \frac{3}{4}$ $\rightarrow$$(c) \text{ OM} = \frac{1}{4} a + \frac{3}{4} b$$\sim \quad \quad \sim$$\rightarrow$$\text{ON} = \frac{1}{2} a + \frac{3}{4} b$$\sim \quad \quad \sim$$\rightarrow \quad \quad \rightarrow$$\text{OM} = \frac{1}{2} \text{ON}$ Hence parallel, and point O is common $\Rightarrow$ O, M and N are collinear M1 M1 M1 B1 B1 B1 B1																																																																																					
19	<table><tr><td>class</td><td>x</td><td>f</td><td>d=x-a</td><td>df</td><td>df²</td><td>cf</td></tr><tr><td>1-10</td><td>5.5</td><td>2</td><td>-40</td><td>-8</td><td>3200</td><td>2</td></tr><tr><td>11-20</td><td>15.5</td><td>6</td><td>-30</td><td>-180</td><td>5400</td><td>8</td></tr><tr><td>21-30</td><td>25.5</td><td>15</td><td>-20</td><td>-300</td><td>6000</td><td>23</td></tr><tr><td>31-40</td><td>35.5</td><td>22</td><td>-10</td><td>-220</td><td>2200</td><td>45</td></tr><tr><td>41-50</td><td>45.5</td><td>36</td><td>0</td><td>0</td><td>0</td><td>81</td></tr><tr><td>51-60</td><td>55.5</td><td>34</td><td>10</td><td>340</td><td>3400</td><td>115</td></tr><tr><td>61-70</td><td>65.5</td><td>20</td><td>20</td><td>400</td><td>8000</td><td>135</td></tr><tr><td>71-80</td><td>75.5</td><td>15</td><td>30</td><td>450</td><td>13500</td><td>150</td></tr><tr><td>81-90</td><td>85.5</td><td>6</td><td>40</td><td>240</td><td>9600</td><td>156</td></tr><tr><td>91-100</td><td>95.5</td><td>4</td><td>50</td><td>200</td><td>10000</td><td>160</td></tr><tr><td></td><td></td><td>160</td><td></td><td>850</td><td>61300</td><td></td></tr></table>	class	x	f	d=x-a	df	df ²	cf	1-10	5.5	2	-40	-8	3200	2	11-20	15.5	6	-30	-180	5400	8	21-30	25.5	15	-20	-300	6000	23	31-40	35.5	22	-10	-220	2200	45	41-50	45.5	36	0	0	0	81	51-60	55.5	34	10	340	3400	115	61-70	65.5	20	20	400	8000	135	71-80	75.5	15	30	450	13500	150	81-90	85.5	6	40	240	9600	156	91-100	95.5	4	50	200	10000	160			160		850	61300		B1 B1 B1 B1 M For d For df For df² For cf
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	$\text{Mean} = 45.5 + \frac{850}{160} = 50.8125$ $\text{Standard deviation} = \sqrt{\frac{6130}{160} - \left(\frac{850}{160}\right)^2}$ $= 18.84$) 121st student $60.5 + \frac{121 - 115}{20} \times 10$ $= 60.5 + 3$ $= 63.5$	1 A1 M 1 A1 M 1 A1	
20	$(a) \begin{pmatrix} m \\ 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} A & B & C & D \\ 3 & 2 & 4 & 5 \\ 1 & 4 & 3 & 1 \end{pmatrix} = \begin{pmatrix} A1 & B1 & C1 & D1 \\ -1 & -4 & -3 & -1 \\ 3 & 2 & 4 & 5 \end{pmatrix}$ $\Rightarrow A^1(-1, 3), B^1(-4, 2), C^1(-3, 4), D^1(-1, 5).$ (i) This is $\frac{1}{4}$ turn about the origin $(b) \begin{pmatrix} N \\ 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} A1 & B1 & C1 & D1 \\ -1 & -4 & -3 & -1 \end{pmatrix} = \begin{pmatrix} A^{11} & B^{11} & C^{11} & D^{11} \\ -1 & -4 & -3 & -1 \\ -3 & -2 & -4 & -5 \end{pmatrix}$ $A^{11}(-1, -3), B^{11}(-4, -2), C^{11}(-3, -4), D^{11}(-1, -5).$ (c) $\begin{pmatrix} N \\ 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} M \\ 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ $\text{Det} = 0 - (+1) = -1$ $\text{Inverse Matrix} = \frac{1}{-1} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$ This is reflection on the line $y = -x$	B1 B1 M 1 A1 B1 B1 B1	
21	(a)	B1 B1 B1	Diagram positions



B1

B1

B1

B1

B1

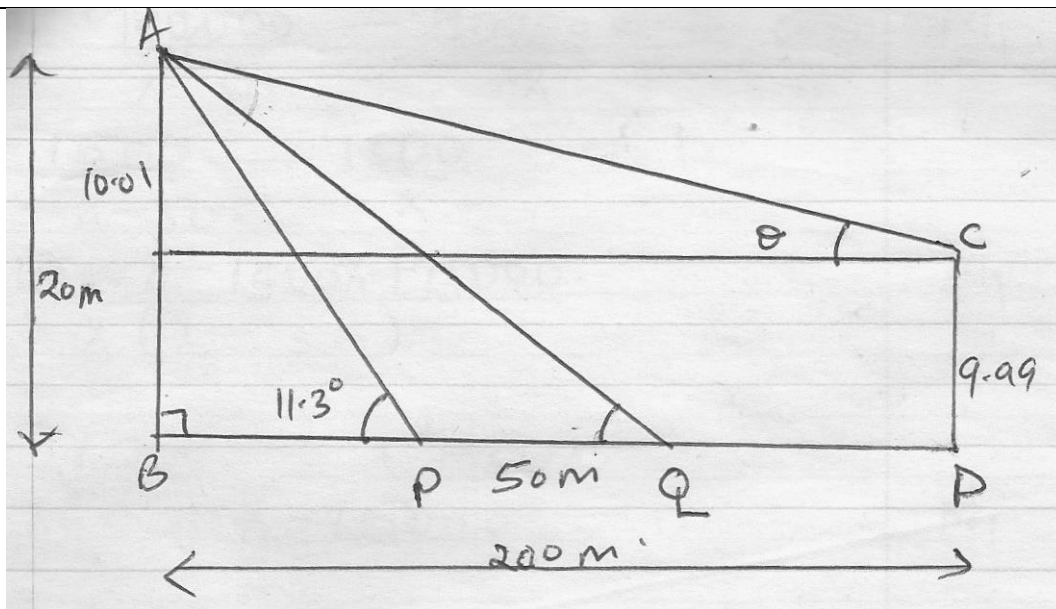
- (b)(i) The distance PS = 11.9 cm (3570 km.)
- (ii) The distance RS = 12.8 cm (3840 km)
- (iii) The bearing of Town S from Q
 $= (270 + 42)$
 $= 312^\circ$ or N 48° W

22

M
1

A1

M
1

M
1

A1

M
1A1
M
1M
1

A1

B1

$$(a) \tan 11.3^\circ = \frac{20}{BP}$$

$$\therefore BP = \frac{20}{\tan 11.3^\circ} = 100.09m$$

$$(b) \text{Time Taken in hrs} = \frac{5}{3600}$$

Distance between P and Q

$$= \left(\frac{36}{1} \times \frac{5}{3600} \right)$$

$$= 0.05km$$

$$50km$$

$$\Rightarrow \tan \alpha = \frac{20}{150.09}$$

$$\alpha = \tan^{-1} \left(\frac{20}{150.09} \right) = 7.59^\circ$$

$$(c) (i) CD^2 = 50.9^2 - 49.91^2 = 99.8019$$

$$CD = \sqrt{99.8019} = 9.9901$$

$$= 9.99m$$

$$(ii) \tan \alpha = \frac{10.01}{200} = 0.05005$$

$$\alpha = \tan^{-1} (0.05005)$$



	$= 2.865$ $= 3.00$		
23.	$\frac{1500}{x-50} - \frac{1500}{x} = 1$ $\frac{1500x - 1500x + 75000}{x(x-50)}$ $75000 = x^2 - 50x$ $x^2 - 50x - 75000 = 0$ $x^2 - 300x + 250x - 75000 = 0$ $x(x-300) + 250(x-300) = 0$ $(x-300)(x+250) = 0$ $x = 300 \text{ or } -250$ $\Rightarrow x = 300$ $(b) \frac{900,000}{250} - \frac{900,000}{300} \times 100\%$ $\frac{900,000}{300} = \left(\frac{3600 - 3000}{300} \right) \times 100\%$ $= 20\%$ $(c) \text{ B:G}$ $11:7$ $\frac{11}{18} \times 900,000$ $\text{Kshs. } 550,000$	M 1 M 1 M 1 M 1 A1 M 1 M 1 A1 M 1 A1 10	
24.	$(a) S = t^3 - 6t^2 + 9t + 5$ $\frac{ds}{dt} = 3t^2 - 12t + 9$ At t = 0.5 seconds. Gradient = $3(0.5)^2 - 12(0.5) + 9$	M 1 M	



	$=3.75\text{m}$	1	
	(b) when $\frac{ds}{dt} = 0, 3t^2 - 12t + 9 = 0$	A1	
	$t^2 - 4t + 3 = 0$		
	$(t-1)(t-3) = 0$		
	$t = 1$ or 3 seconds	M	
	when $t=1\text{s}$, $S=1-6+9+5$	1	
	$=9\text{m}$		
	When $t = 3\text{s}$, $S=3^3 - 6 \times 3^3 + 9 \times 3 + 5$	A1	
	$=5\text{m}.$		
		A1	
	(c) $\frac{d^2s}{dt^2} = 6t - 12$		
	At $t = 1\text{s}$, $\frac{d^2s}{dt^2} = 6 \times 1 - 12 = -6$		
	$\therefore$ at $(1,9)$ max turning point	M	
		1	
	At $t = 3\text{s}$, $\frac{d^2s}{dt^2} = 18 - 12 = 6$	M	
		1	
	At $(3,5)$ Min turning point.		
		A1	
		10	

